

SCORIFY

Phase 4: Model Evaluation & Detailed Analysis

F2602

1. Evaluation Metrics

In Phase 3 we trained three models on both targets. Here we do a proper breakdown of what the numbers actually mean for the business.

Target 1 — Conversion Probability (XGBoost, best model):

Metric	Value	What it means
MAE	0.0227	On average the model is off by 2.3%. If it says 15% conversion, actual is between 12.7% and 17.3%.
RMSE	0.0454	RMSE is double the MAE which means there are some outlier predictions where the model misses badly.
R ²	0.7159	The model explains 71.6% of why some leads convert and others don't. Remaining 28.4% is stuff we can't capture from these features.

Target 2 — Profit Prediction (Random Forest, best model):

Metric	Value	What it means
MAE	\$20,774	On average the model is off by about \$20K. If it predicts \$100K profit, actual is likely between \$80K and \$120K.
RMSE	\$32,447	Higher than MAE, meaning some deals have much larger errors, especially premium deals above \$300K.
R ²	0.7608	The model captures 76.1% of what drives profit differences between leads.

2. Comparison of Models

We compared three models on both targets. Results on the test set (492 leads):

Conversion Probability:

Model	MAE	RMSE	R ²
Linear Regression	0.0386	0.0652	0.4130
Random Forest	0.0238	0.0468	0.6977
XGBoost	0.0228	0.0450	0.7202

Profit Prediction:

Model	MAE	RMSE	R ²
Linear Regression	32,300	42,735	0.5851

Random Forest	20,726	32,584	0.7588
XGBoost	20,719	34,697	0.7265

Linear Regression scored the lowest on everything, which confirms the data has non-linear patterns. A straight line simply cannot capture the complex relationship between engagement signals and conversion/profit. XGBoost won for conversion probability while Random Forest won for profit prediction. Both tree-based models were significantly better than Linear Regression.

3. Error Analysis

We looked at where the model performs well and where it struggles by grouping the test leads into ranges:

Conversion — Error by actual probability range:

Range	Count	Avg Error	Observation
Very Low (0-2%)	247	0.005	Very accurate
Low (2-5%)	94	0.020	Good
Medium (5-10%)	71	0.028	Acceptable
High (10-20%)	47	0.055	Struggles
Very High (20%+)	33	0.105	Poor

The pattern is clear: the model is great at predicting cold leads (0-5% range) because there are plenty of them in the training data. But when it comes to hot leads (20%+), the average error jumps to 10.5%. This happens because only about 7% of all leads have conversion probability above 10%, so the model has fewer examples to learn from in that range.

Profit — Error by deal size:

Range	Count	Avg Error	% Error
Low (<50K)	23	\$18,419	43.1%
Medium (50-100K)	255	\$11,483	16.5%
High (100-150K)	108	\$21,967	18.0%
Very High (150-300K)	95	\$40,084	21.1%
Premium (300K+)	11	\$62,577	15.7%

The model is most accurate for medium deals (50-100K) with just 16.5% error. The lowest deals (<50K) have the highest percentage error at 43% — these are unusual in the dataset and the model doesn't have enough examples to learn their patterns. Premium deals (300K+) have high absolute error (\$62K) but actually decent percentage error (15.7%).

Overprediction vs underprediction: the model is balanced — 50.4% overpredictions and 49.6% underpredictions for conversion. It doesn't systematically lean one way.

Worst predictions: the biggest conversion miss was an actual 59.4% lead predicted at 21.2% (off by 38 points). For profit, the worst was an actual \$669K deal predicted at \$447K (off by \$222K). Both worst cases involve extreme values that are rare in the training data.

4. Bias / Fairness Check

We checked whether the model performs equally across different lead groups or if it favours certain segments:

By Device:

Device	Count	Conv MAE	Profit MAE
Mobile	283	0.013	\$20,880
Desktop	159	0.036	\$18,209
Unknown	42	0.036	\$30,990

Interesting finding: mobile leads actually get more accurate conversion predictions (MAE 0.013) than desktop (0.036). This is likely because mobile leads make up 57% of the data so the model has more examples. For profit prediction, desktop leads are slightly better. The "Unknown" device group has the worst predictions across the board, which makes sense — less information means less accurate predictions.

By Travel Group Type:

Group	Count	Conv MAE	Profit MAE
Partner	195	0.024	\$16,395
Family	130	0.025	\$21,907
Friends	66	0.016	\$22,675
Solo	62	0.013	\$22,391
Unknown	39	0.037	\$33,101

No major fairness concerns across travel types. Solo and Friends leads get the most accurate conversion predictions. Partner leads get the best profit accuracy (\$16K error). The "Unknown" group performs worst again — this is expected and not a bias issue, it is a data completeness issue.

By Budget Level:

Budget	Count	Conv MAE	Profit MAE
Low	220	0.018	\$16,537
Not Sure	92	0.016	\$26,252
Medium	83	0.036	\$18,156
High	58	0.025	\$23,609

Low budget leads get the best predictions overall. Medium budget leads have higher conversion error (0.036) which might be because they are the hardest group to predict — some of them end up converting while others drop off, and the model has a harder time telling them apart.

5. Key Insights and Observations

- **Tree models beat linear every time.** XGBoost and Random Forest both outperformed Linear Regression by a huge margin on both targets. This confirms the data has non-linear patterns that a straight line simply cannot capture.
- **The model is best at identifying cold leads.** It predicts low-conversion leads (0-5%) very accurately with just 0.5-2% error. This is actually the most useful outcome for the business — it confidently tells you which leads NOT to waste time on.
- **Hot leads are harder to predict.** The 20%+ conversion range has the highest error. This is a data imbalance problem — only 33 out of 492 test leads fall in this range. More data from high-converting leads would fix this.
- **Engagement features matter more than profile data.** time_on_site, total_sessions, enquiry_touchpoints, and page_interactions are the top predictors. Budget level and travel type help but are secondary. A lead who spends 15 minutes browsing and fills 3 forms is more predictive than one who says "high budget" but only visits once.
- **No major bias issues.** The model performs consistently across devices, travel types, and budget levels. The only weak spots are groups with missing data ("Unknown" categories) which is expected.

- **Medium deals are the sweet spot.** Profit predictions are most accurate for deals between 50-100K, which happens to be where most of Go2Africa's business sits.

6. Challenges Faced

- **Imbalanced target distribution** — most leads have very low conversion probability (mean 5%, median 2%). The model learns the common pattern well but struggles with the rare high-conversion leads. More balanced training data or oversampling techniques could help.
- **Evaluating with small group sizes** — some groups in the bias check had very few samples (Italy with 12 leads, Premium deals with 11). It is hard to draw reliable conclusions from such small groups. We only reported results for groups with 10+ samples.
- **Negative R^2 within narrow ranges** — when we calculated R^2 within a specific conversion range (like just the 2-5% leads), R^2 was negative. This does not mean the model is terrible — it means within a very narrow band the model's variation is less useful than just predicting the group mean. Overall R^2 across the full range remains strong at 0.72.
- **No external validation** — we evaluated on a held-out test set from the same dataset. We have not yet tested on leads from a completely different time period or a different company. Real-world deployment would need this kind of validation.